

technologies that appeared in 2002. After Term Weighting, 'DDR RAM' would have a higher value under the column of the former document.

The next step is Normalisation; this ensures that large documents with a lot of keywords do not override smaller, but potentially more important, documents. The values under the columns of larger documents are reduced to give every document a fair chance. Now consider that in the matrix, generally, a certain keyword would appear only in a few documents, and therefore the resulting matrix would be sparse—a matrix with most of the elements being 0s. Using a mathematical procedure called Singular Value Decomposition (SVD), the matrix is compressed so that it has fewer zeroes. By this process, keywords with semantically closer meanings come together. Using this index with supplied keywords, we can retrieve documents having a value higher than a critical value.

LSI indexes need to be modified every time a document is added; to worsen things, the SVD step is computation-intensive, and in the end, a 7,000 docu-

ment collection takes about 40 minutes to index. This makes LSI out of bounds in its current form for Web search engines, but LSI can be used for searching through personal documents, or a fixed but large number of documents. For instance, LSI is used in the Mac OS 10.2 spam filter.

Contextual Network Graphs (CNG) is an alternative to LSI, and uses graphs



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// ... the fact of the matter is that there really hasn't been much progress in the basic science of how to search since the seventies. ... We're not likely to get an intelligent search engine anytime soon //

instead of matrices. In CNG, documents and keywords are represented in an interlinked graph, with documents and keywords as nodes. This makes it easier to scale, and is not as computationally intensive as LSI. Here again, the problem is that CNG is a relatively new technique, and can't handle long queries like LSI can.

Searching is a difficult proposition even with current hard disks. Imagine

Personal Search

What about the data on your own hard disk? If you have an 80 GB or even a 40 GB hard disk, you know how difficult it is to find a particular file, or a file containing a particular snippet of text. Search engines such as Sleuthhound (www.pcworld.com/downloads) are available for searching your hard disk. An expanded version of this idea is Atlantis InterNetworks' Comet Data Engine (www.atlinet.com), a combination of a Web crawler and a search engine, for use with Windows. The data engine is used to index and search HTML and other text documents on the intranet and on the Internet. It is composed of three component engine—the Acquisition, Indexing and Retrieval engines. These components work together to provide high speed indexing and search services—essentially, working to minimise the effort of searching for any piece of data, wherever it may be located. Software like these are mostly just improvements upon current search capabilities.

the situation in a few years, when you'll probably have an 800 GB hard disk, containing your entire media collection, along with other data. Can you trust existing search tools to perform a search on so much data? Microsoft has recognised this problem, and is reportedly working on a new file system, dubbed WinFS, that will help users get better search results, both offline and online. The file system allows anything that you've seen on your PC to be retrieved by the local search engine later. This means that anything that you do on the system—online activities such as reading e-mail, or browsing the Web, and offline activities such as working with local files—will be monitored and indexed by the OS. With Longhorn—the upcoming version of Windows—there should no difference between searching offline and online. One of the methods of achieving this is continuous indexing. Indexing is a feature that was introduced in NTFS to enable quicker searches. A lot of radical changes are expected in Longhorn, but nothing concrete has emerged yet.

Another feature that is expected

Search Needs	
While most of us may be satisfied with current search engines, there is a lot that has to be done and that can be done	
Current Problems and Requirements	Solutions
Current search engines look for exact word matches, not semantic matches	Semantic search techniques such as LSI and CNS, and related techniques such as NLP, can give meaningful results
Search results should be presented in a way that allows for easier identification of results	Presenting the results in a visual or more useful way, say, grouping results like Vivismo does
Organising and searching local files is tedious	Continuous indexing and use of query-based virtual folders
Need for alternate methods of entering a search query, like voice search to search from mobile phones	Voice-based search such as the one being tested by Google labs
Ability to retrieve unstructured information like unannotated images and videos	Image recognition techniques to be used with search techniques
No personalised search tool that searches in many locations such as local disk, Internet, Inbox	Search systems that index multiple locations; expected to appear in Longhorn